

Assignment 1

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Introduction

National Supported Work (NSW) Demonstration was a subsidized job training program spanning 1974 to 1979 designed to help disadvantaged workers including long-term AFDC (welfare) recipients, ex-addicts, ex-offenders, and young school dropouts secure stable employment (Manpower Demonstration Research Corporation, 1983). Historically, econometric evaluations of such programs faced immense confounding and consequently produced unreliable causal inferences. This paper attempts to estimate the effect of program enrollment on real trainee earnings where participants were randomly assigned to treatment (job training) and control groups (no training). By applying the Causal Roadmap to the classic LaLonde dataset (1986), we can follow the Causal Roadmap to define our assumptions and compare three statistical estimators: G-computation (standardization), Augmented Inverse Probability of Treatment Weighting (AIPTW), and Targeted Maximum Likelihood Estimation (TMLE) with Ensemble SuperLearner (SL). More information on the LaLonde dataset can be found in the documentation for the R package ‘designmatch’.

Research Question

We are interested in knowing to what extent participation in the NSW Demonstration program affects future income among Black participants. In causal phrasing: if, perhaps contrary to reality, all eligible Black individuals received the NSW Demonstration training program, what would their expected real earnings in 1978 be compared to if, perhaps contrary to reality, none of them received the program?

Data

To answer this question, we observe the data structure $O = (W, A, Y)$ for n independent and identically distributed individuals.

- W is a set of 6 baseline covariates that could confound the effect of A on Y : age, years of education, marital status, absence of a high school degree, and real earnings in 1974 and 1975.
- $A \in \{0, 1\}$ is the binary intervention. $A = 1$ if the individual received the job training, and $A = 0$ otherwise.
- Y is the continuous outcome of interest: real earnings in 1978.

Our sample is comprised of 243 Black individuals, corresponding to 156 treated and 87 control subjects, and 10 the aforementioned variables.

```
data(lalonde)

lalonde <- subset(lalonde, black == 1) # Only Black people for the ATE

summary(lalonde) # Everything looks good, no missingness
```

```
##      treatment      age      education      black      hispanic
## Min.   :0.000   Min.   :16.00   Min.    : 1.00   Min.    :1   Min.    :0
## 1st Qu.:0.000   1st Qu.:19.00   1st Qu.: 9.00   1st Qu.:1   1st Qu.:0
## Median :1.000   Median :24.00   Median :11.00   Median :1   Median :0
## Mean   :0.642   Mean   :26.01   Mean    :10.23   Mean    :1   Mean    :0
## 3rd Qu.:1.000   3rd Qu.:29.00   3rd Qu.:12.00   3rd Qu.:1   3rd Qu.:0
## Max.   :1.000   Max.   :55.00   Max.    :17.00   Max.    :1   Max.    :0
##      married      nodegree      re74      re75
## Min.   :0.0000   Min.    :0.0000   Min.    : 0   Min.    : 0
## 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 0   1st Qu.: 0
## Median :0.0000   Median :1.0000   Median : 0   Median : 0
## Mean   :0.2222   Mean    :0.6955   Mean    :2499   Mean    :1614
## 3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:2048   3rd Qu.:2039
## Max.   :1.0000   Max.    :1.0000   Max.    :35040   Max.    :25142
##      re78
## Min.    : 0
## 1st Qu.: 0
## Median :2821
## Mean    :5677
## 3rd Qu.:8984
## Max.    :60308
```

```
lalonge <- lalonge |>
  select(-hispanic, -black)

summary(lalonge) # Everything looks good, no missingness
```

```
##      treatment      age      education      married
## Min.   :0.000   Min.   :16.00   Min.    : 1.00   Min.    :0.0000
## 1st Qu.:0.000   1st Qu.:19.00   1st Qu.: 9.00   1st Qu.:0.0000
## Median :1.000   Median :24.00   Median :11.00   Median :0.0000
## Mean   :0.642   Mean   :26.01   Mean    :10.23   Mean    :0.2222
## 3rd Qu.:1.000   3rd Qu.:29.00   3rd Qu.:12.00   3rd Qu.:0.0000
## Max.   :1.000   Max.   :55.00   Max.    :17.00   Max.    :1.0000
##      nodegree      re74      re75      re78
## Min.   :0.0000   Min.    : 0   Min.    : 0   Min.    : 0
## 1st Qu.:0.0000   1st Qu.: 0   1st Qu.: 0   1st Qu.: 0
## Median :1.0000   Median : 0   Median : 0   Median :2821
## Mean   :0.6955   Mean    :2499   Mean    :1614   Mean    :5677
## 3rd Qu.:1.0000   3rd Qu.:2048   3rd Qu.:2039   3rd Qu.:8984
## Max.   :1.0000   Max.    :35040   Max.    :25142   Max.    :60308
```

```
# Define Covariates (W), Treatment (A), and Outcome (Y)
W <- lalonge[, c("age", "education",
                 "married", "nodegree", "re74", "re75")]
A <- lalonge$treatment # 1 = treated; 0 = control
Y <- lalonge$re78 # 1978 earnings
```

Causal Estimand and Identification

Our target parameter is the Average Treatment Effect (ATE), defined under counterfactual outcomes as $E[Y^{a=1} - Y^{a=0} | Black = 1]$. This represents the difference in expected 1978 earnings among Black people where $Y^1 | Black = 1$ is the counterfactual outcome if one, perhaps contrary to reality, recieved job training through NSW and $Y^0 | Black = 1$ is the counterfactual outcome if one, perhaps contrary to reality, did not

receive job training (control).

To identify the causal ATE from our observational data, we must meet three assumptions:

- Consistency: An individual's observed outcome Y when they receive treatment $A = a$ is equal to their counterfactual outcome Y^a .
- Marginal Exchangeability: $Y^a \perp A$. All common causes of the job training assignment and 1978 earnings are controlled for by random treatment assignment.
- Positivity: $0 < p(A = 1 \mid W) < 1$ for observed strata W . Every individual must have a non-zero probability of receiving or not receiving the training.

Consider $E[Y^1]$:

- $E[Y^1 \mid \text{Black} = 1] = E[E[Y^1 \mid \text{Black} = 1]]$ (by the law of iterated expectations)
- $= E[E[Y^1 \mid A = 1, \text{Black} = 1]]$ (by marginal exchangeability, $Y^a \perp A$)
- $= E[E[Y \mid A = 1, \text{Black} = 1]]$ (by consistency)

We can replace the 1's with 0's to find $E[Y^0 \mid \text{Black} = 1]$. Thus, the causal ATE is identified by the statistical estimand:

$$\hat{ATE} = \mathbb{E}[\mathbb{E}[Y \mid A = 1, \text{Black} = 1] - \mathbb{E}[Y \mid A = 0, \text{Black} = 1]]$$

Estimating the Average Treatment Effect

We apply three estimators discussed in class to determine estimate the statistical estimand: g-computation (standardization), augmented inverse-propensity of treatment weights (AIPTW), and TMLE with Ensemble SL (van der Laan et al., 2007). AIPTW is doubly-robust, so naturally it should leverage the predicted values from both the outcome model (calculated in the G-computation step) and the propensity score model. Ensemble SL yields flexible estimation of the propensity score and outcome regression, and the doubly-robust TMLE estimator ensures favorable asymptotic properties as long as one of these (propensity score or outcome regression) is correctly specified. Both of these methods are known to be more statistically powerful than G-computation alone.

```
# Fit the outcome model predicting Y from A and W
gcomp_model <- glm(re78 ~ treatment + age + education + married + nodegree + re74 + re75, data = lalonde)

# Create two counterfactual datasets
data_A1 <- lalonde
data_A1$treatment <- 1 # All treated

data_A0 <- lalonde
data_A0$treatment <- 0 # All untreated

# Predict the counterfactual outcomes
QAW_1 <- predict(gcomp_model, newdata = data_A1, type = "response") # All treated counterfactual outcomes
QAW_0 <- predict(gcomp_model, newdata = data_A0, type = "response") # All untreated counterfactual outcomes

# Calculate the ATE
ate_gcomp <- mean(QAW_1 - QAW_0) # ATE

# Calculate SE for G-comp using a non-parametric bootstrap (e.g., 200 iterations)
set.seed(123)
B <- 200
n_obs <- nrow(lalonde)
boot_ates <- numeric(B)
```

```

for(i in 1:B) {
  # Sample with replacement
  boot_indices <- sample(1:n_obs, replace = TRUE)
  boot_data <- lalonde[boot_indices, ]

  # Refit the model on the bootstrap sample
  boot_model <- glm(re78 ~ treatment + age + education
                    + married + nodegree + re74 + re75, data = boot_data, family = gaussian())

  # Create counterfactuals for the bootstrap sample
  boot_A1 <- boot_data; boot_A1$treatment <- 1 # All treated
  boot_A0 <- boot_data; boot_A0$treatment <- 0 # All untreated

  # Predict and calculate the bootstrap RD for get the ATE
  boot_Q1 <- predict(boot_model, newdata = boot_A1, type = "response")
  boot_Q0 <- predict(boot_model, newdata = boot_A0, type = "response")
  boot_ates[i] <- mean(boot_Q1 - boot_Q0)
}

# The SE is the standard deviation of the bootstrap ATEs
se_gcomp <- sd(boot_ates)

# CI for G-computation
lower_gcomp <- ate_gcomp - 1.96 * se_gcomp
upper_gcomp <- ate_gcomp + 1.96 * se_gcomp
ci_gcomp <- paste0("(", round(lower_gcomp, 2), ", ", round(upper_gcomp, 2), ")")

### Estimator 2: Augmented Inverse Probability of Treatment Weighting (AIPTW)

# Fit the propensity score model (g-model)
ps_model <- glm(treatment ~ age + education
                + married + nodegree + re74 + re75,
                data = lalonde, family = binomial())

# Get the predicted probabilities
g1 <- predict(ps_model, type = "response") # Predicted probability of being treated for each individual
g1 <- pmax(0.025, pmin(0.975, g1)) # Truncate for better positivity
g0 <- 1 - g1 # Predicted probability of being untreated for each individual

# Calculate the AIPTW components for the counterfactual means
Y1_aiptw <- (lalonde$treatment / g1) * (lalonde$re78 - QAW_1) + QAW_1
Y0_aiptw <- ((1 - lalonde$treatment) / g0) * (lalonde$re78 - QAW_0) + QAW_0

# Calculate the ATE
ate_aiptw <- mean(Y1_aiptw - Y0_aiptw)

# Calculate SE for AIPTW using the variance of the influence curve
IC_aiptw <- (Y1_aiptw - Y0_aiptw) - ate_aiptw
var_aiptw <- var(IC_aiptw) / n_obs
se_aiptw <- sqrt(var_aiptw)

# CI for AIPTW
lower_aiptw <- ate_aiptw - 1.96 * se_aiptw

```

Table 1: ATE Estimates on LaLonde Data

Estimator	ATE	CI
G-computation	1371.45	(-281.69, 3024.59)
AIPTW	1417.46	(-408.16, 3243.09)
TMLE	873.43	(-582.24, 2329.11)

```

upper_aiptw <- ate_aiptw + 1.96 * se_aiptw
ci_aiptw <- paste0("(", round(lower_aiptw, 2), ", ", round(upper_aiptw, 2), ")")

### Estimator 3: TMLE with Ensemble SuperLearner (SL)

# Reorder the dataset chronologically since LTMLE will get mad at me: Covariates -> Treatment -> Outcome
lalonge <- lalonge[, c("age", "education", "married",
                      "nodegree", "re74", "re75", "treatment", "re78")]

# Define the SuperLearner library
sl_lib <- c("SL.glm", "SL.mean", "SL.gam", "SL.earth", "SL.xgboost")

# Run TMLE
set.seed(123)
tmle_fit <- suppressWarnings(suppressMessages(ltmle(lalonge,
  Anodes = "treatment", # Treatment
  Ynodes = "re78", # Outcome
  SL.library = sl_lib,
  abar = list(1, 0))))

# Extract the ATE and CI
tmle_summary <- summary(tmle_fit)
ate_tmle <- tmle_summary$effect.measures$ATE$estimate
tmle_ci_bounds <- tmle_summary$effect.measures$ATE$CI
ci_tmle <- paste0("(", round(tmle_ci_bounds[1], 2), ", ", round(tmle_ci_bounds[2], 2), ")")

### Compile all results

# Compile all results into a single data frame
results_df <- data.frame(
  Estimator = c("G-computation", "AIPTW", "TMLE"),
  ATE = c(ate_gcomp, ate_aiptw, ate_tmle),
  CI = c(ci_gcomp, ci_aiptw, ci_tmle)
)

# Output as a table for readability
kable(results_df, format = "latex", digits = 2, caption = "ATE Estimates on LaLonde Data") %>%
  kable_styling(bootstrap_options = c("striped", "hold_position"))

```

G-computation returns a causal $\hat{ATE}$ of \$1371.454 (95% CI: -281.69, 3024.59), AIPTW returns a causal $\hat{ATE}$ of \$1417.464 (95% CI: -408.16, 3243.09), and TMLE + SL returns a causal $\hat{ATE}$ of \$873.432 (95% CI: -582.24, 2329.11).

If unmeasured confounding, marginal exchangeability, consistency and positivity hold, this implies that among Black people, enrollment in the NSW Demonstration increased real earnings in 1978 by more than those who weren't enrolled in the program.

Conclusion

Under consistency, marginal exchangeability, and positivity, this study successfully identified the causal $\hat{ATE}$ for the effect of NSW Demonstration enrollment on real earnings in 1978 among a small sample of Black individuals. G-computation, AIPTW, and TMLE + SL appear to show increases in earnings with program enrollment for Black individuals, but this finding may or may not be due to chance. The wide confidence interval may be due to low predicted treatment probabilities for certain covariate combinations, or the fact that we do not have access to unmeasured confounders of the effect.

References:

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